

Overview of Contrastive learning

Subject: Contrastive learning

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(z_i, z_i)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

1.

Non-contrastive self-supervised learning Non-contrastive self-supervised learning (NCSSL) uses only positive examples. Counterintuitively, NCSSL converges on a useful local minimum rather than reaching a trivial solution, with zero loss. For the example of binary classification, it would trivially learn to classify each example as positive. Effective NCSSL requires an extra predictor on the online side that does not back-propagate on the target side.

2.

Self-supervised learning (SSL) is a paradigm in machine learning where a model is trained on a task using the data itself to generate supervisory signals, rather than relying on externally-provided labels. In the context of neural networks, self-supervised learning aims to leverage inherent structures or relationships within the input data to create meaningful training signals. SSL tasks are designed so that solving them requires capturing essential features or relationships in the data. The input data is typically augmented or transformed in a way that creates pairs of related samples, where one sample serves as the input, and the other is used to formulate the supervisory signal. This augmentation can involve introducing noise, cropping, rotation, or other transformations. Self-supervised learning more closely imitates the way humans learn to classify objects. During SSL, the model learns in two steps. First, the task is solved based on an auxiliary or pretext classification task using pseudo-labels, which help to initialize the model parameters. Next, the actual task is performed with supervised or unsupervised learning. Self-supervised learning has produced promising results in recent years, and has found practical application in fields such as audio processing, and is being used by Facebook and others for speech recognition.

3.

Joint-Embedding and Predictive Architectures A major class of self-supervised learning moves beyond contrastive pairs, instead maximizing the agreement between views while preventing collapse through statistical constraints. Rooted in Deep Canonical Correlation Analysis (Deep CCA), this approach includes Joint-Embedding Architectures (JEA) like Barlow Twins and VICReg, which enforce covariance constraints to learn invariant representations without negative sampling. Deep Latent Variable Path Modelling (DLVPM) generalizes this to multimodal systems, using path models to enforce correlation and orthogonality across diverse data types. In 2022, this framework evolved into Joint-Embedding Predictive Architectures (JEPA) when Yann LeCun introduced JEPA as a step towards decision making, reasoning, and autonomous human intelligence in machines, including self-improvement through autonomous learning. Founded in representation learning, LeCun established the concept of a “world model” in JEPA which aims to enable machines to replicate human

intellect by providing machines with a concept for the world in which they exist. Unlike autoencoders, JEPAs operate entirely in latent space, avoiding pixel-level noise to focus on semantic structure, a key step toward autonomous world models. Rather than just learning invariance, JEPAs learn by predicting masked latent representations from visible context. Recently, there have been several new applications of JEPA for things like image analysis, audio processing, and even motion in images and video.

4.

Examples Self-supervised learning is particularly suitable for speech recognition. For example, Facebook developed wav2vec, a self-supervised algorithm, to perform speech recognition using two deep convolutional neural networks that build on each other. Google's Bidirectional Encoder Representations from Transformers (BERT) model is used to better understand the context of search queries. OpenAI's GPT-3 is an autoregressive language model that can be used in language processing. It can be used to translate texts or answer questions, among other things. Bootstrap Your Own Latent (BYOL) is a NCSSL that produced excellent results on ImageNet and on transfer and semi-supervised benchmarks. The Yarowsky algorithm is an example of self-supervised learning in natural language processing. From a small number of labeled examples, it learns to predict which word sense of a polysemous word is being used at a given point in text. DirectPred is a NCSSL that directly sets the predictor weights instead of learning it via typical gradient descent. Self-GenomeNet is an example of self-supervised learning in genomics. Self-supervised learning continues to gain prominence as a new approach across diverse fields. Its ability to leverage unlabeled data effectively opens new possibilities for advancement in machine learning, especially in data-driven application domains.

5.

Comparison with other forms of machine learning SSL belongs to supervised learning methods insofar as the goal is to generate a classified output from the input. At the same time, however, it does not require the explicit use of labeled input-output pairs. Instead, correlations, metadata embedded in the data, or domain knowledge present in the input are implicitly and autonomously extracted from the data. These supervisory signals, extracted from the data, can then be used for training. SSL is similar to unsupervised learning in that it does not require labels in the sample data. Unlike unsupervised learning, however, learning is not done using inherent data structures. Semi-supervised learning combines supervised and unsupervised learning, requiring only a small portion of the learning data be labeled. In transfer learning, a model designed for one task is reused on a different task. Training an autoencoder intrinsically constitutes a self-supervised process, because the output pattern needs to become an optimal reconstruction of the input pattern itself. However, in current jargon, the term 'self-supervised' often refers to tasks based on a pretext-task training setup. This involves the (human) design of such pretext task(s), unlike the case of fully self-contained autoencoder training. In reinforcement learning, self-supervising learning from a combination of losses can create abstract representations where only the most important information about the state are kept in a compressed way.

6.

Contrastive self-supervised learning For a binary classification task, training data can be divided into positive examples and negative examples. Positive examples are those that match the target. For example, if training a classifier to identify birds, the positive training data would include images that contain birds. Negative examples would be images that do not. Contrastive self-supervised learning

uses both positive and negative examples. The loss function in contrastive learning is used to minimize the distance between positive sample pairs, while maximizing the distance between negative sample pairs. An early example uses a pair of 1-dimensional convolutional neural networks to process a pair of images and maximize their agreement. Contrastive Language-Image Pre-training (CLIP) allows joint pretraining of a text encoder and an image encoder, such that a matching image-text pair have image encoding vector and text encoding vector that span a small angle (having a large cosine similarity). InfoNCE (Noise-Contrastive Estimation) is a method to optimize two models jointly, based on Noise Contrastive Estimation (NCE). Given a set $X = \{x_1, \dots, x_N\}$ of N random samples containing one positive sample from $p(x_{t+k} \mid c_t)$ and $N - 1$ negative samples from the 'proposal' distribution $p(x_{t+k})$, it minimizes the following loss function: